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EXTENDED-ABSTRACT

SlateLLM: Distilling LLM Semantics into Session-Aware Slate Recommendation without Inference Overhead

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SlateLLM: Distilling LLM Semantics into Session-Aware Slate Recommendation without Inference Overhead

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Abstract

Session-based slate recommendation systems curate ranked sets of items in real-time, adapting to evolving user interactions. Balancing relevance, diversity, and novelty remains challenging for reinforcement learning (RL) methods. Recent advances in large language models (LLMs) offer a new possibility to leverage their semantic reasoning capabilities to refine slate composition. In this work, we examine the impact of LLM-driven reasoning on slate generation by integrating LLMs with an RL-based slate recommender and evaluating in terms of accuracy, similarity, diversity, and novelty. We extend the RecSim framework with real-world interaction data and introduce a session-aware evaluation protocol that captures long-term engagement. Our analysis reveals that LLM reasoning enhances subcategory-level diversity while maintaining relevance, leading to increased user engagement. By visualizing category-level shifts in slate composition we uncover systematic patterns in how LLMs refine recommendation diversity. Although direct LLM use during inference may be hampered by computational demands and latency concerns, our experimental results demonstrate that integrating LLM modifications during training enables the model to internalize the nuanced characteristics of LLM reasoning without incurring inference overhead, thereby improving recommendation performance, serving time efficiency, and deployability.

CCS Concepts

• **Computing methodologies → Sequential decision making**.

Keywords

LLM Reasoning, Slate recommendation, Reinforcement learning, User diversity

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1 Introduction & Related Works

Session-based slate recommendation has emerged as a critical frontier in personalized recommendation systems, addressing the dynamic and transient nature of user interactions within bounded temporal contexts [3, 8, 12, 15]. Unlike traditional sequential recommendation, which predicts the next single item in a user's interaction sequence, session-based slate recommendation focuses on curating *ranked sets of items* (slates) that adapt in real time to a user's evolving behavior. This paradigm is inherently interactive: A user may click on a single item from the slate, triggering an updated set of recommendations, or disengage entirely, prompting the system to recalibrate the next slate. For example, in a *news browsing session*, a user might scroll through a grid of article headlines, click on one, and immediately receive a refreshed slate of related stories; similarly, in a *YouTube video session*, a viewer's choice to watch (or skip) a recommended video dictates the composition of subsequent slates. These iterative interactions persist until the session terminates, making the problem uniquely challenging due to its dependency on *real-time feedback loops* and *combinatorial decision-making*.

Recent advances in this field include reinforcement learning frameworks that model sequential slate-user interactions [1, 8], neural re-ranking architectures that refine candidate sets [15], and hybrid methods incorporating item metadata (e.g., categories, embeddings) to enhance slate diversity and coherence [13, 14]. The influence of NLP research and reasoning in Large Language Models (LLMs), which can further enable these existing methods through their semantically rich reasoning capabilities. Systems leveraging generative pretrained transformers [5] dynamically interpret session context to generate adaptive slates with improved alignment to evolving user intent. Current methods often rely on costly live experiments [7] or purely simulated environments like *RecSim* [16], which suffer from ecological validity limitations [2].

To address these challenges, we present three interconnected contributions, including, to the best of our knowledge, the first approach to incorporate LLM world knowledge into session-based

slate recommendation with reasonable serving time. First, we examine the influence of LLM-based reasoning in existing slate recommendation pipelines. By integrating LLMs like Deepseek-R1 [5] with reinforcement learning agents, we analyze how their semantic understanding enriches slate generation, improving the ability to capture evolving user intent and latent preferences. Second, while using LLMs during inference can be computationally prohibitive, we show that their world knowledge can be distilled into existing RL-based slate recommenders during training. Specifically, we introduce a regularization term applied to the actor network in an existing session based slate framework [14], which allows the recommender to internalize LLM-style semantic reasoning without requiring the LLM at serving time, thus preserving deployability and inference efficiency. Third, we extend the RecSim simulator with real-world interaction logs, enabling a session-aware evaluation protocol that incorporates multi-slate interactions and quantifies long-term engagement effects based on relevance, diversity and novelty metrics.

2 Architecture & Evaluation

Our objective is to how the semantic reasoning capabilities of LLMs can enhance existing RL-based session-based slate recommendation frameworks such as SlateQ [8] and Proto-Slate [14], the latter being an advanced variant of the Proto-Item algorithm [15] designed to support both diverse and focused user preferences. To study this, we first generate an initial slate using these frameworks, and then refine it using an LLM. The LLM is provided with the user history, the original slate, and the set of candidate documents, along with the following structured prompt: *"What changes will you make to this slate? If you think any item needs to be changed, consider selecting it from the candidate docs. Take this decision in such a way that the user's click probability from the slate is maximized. The user is allowed only to pick one item. Provide the recommended slate as a numbered list of 10 items. Keep it short and concise."* The LLM-refined variants of the two frameworks are denoted as Proto-Slate+LLM and SlateQ+LLM, respectively.

To examine whether the observed gains stem from deeper reasoning or superficial matching, we compare these LLM-refined methods against strong similarity-based baselines. Specifically, we create BM25+cosine similarity variants of the original models, where $k = 2$ items in the slate are replaced based on a combined score of title-level BM25 similarity and embedding-level cosine similarity; the value $k = 2$ was chosen as it yielded the best performance across our evaluation metrics. Additionally, we include a baseline where the LLM directly generates a slate given the user's interaction history and the set of candidate documents, allowing us to isolate the effectiveness of LLM reasoning when used without any underlying RL-based slate framework.

While LLM augmentation at inference improves several metrics, it is computationally intensive and unsuitable for real-time deployment. To address this, we incorporate LLM world knowledge into the Proto-Slate framework during training by introducing a regularization term inspired by LLM-generated trajectories [14]. We choose the Proto-Slate framework as its action generation component allows the policy to be trained to align its outputs with LLM-refined actions. This allows the model to learn semantically meaningful slate refinements without relying on LLMs at inference. Although slower training is acceptable, maintaining fast serving is crucial, as latency directly affects user engagement and retention [4].

2.1 LLM-Regularized Proto-Slate Optimization

The existing *Proto-Slate* [13, 14] architecture addresses the problem that slate recommendation entails a very large discrete action space. It maps the problem to a continuous-valued action space and incorporates the deep deterministic policy gradient (DDPG) algorithm within its training algorithm. Write A for a slate (i.e. an ordered set) of $N > 0$ items and $A[i]$ for the ith item on the slate, where $i = 1, \ldots, N$. The action space consists of all possible slates that can be recommended. Discrete items in the system are represented by an m-dimensional embedding and this allows the action-space to be extended to all possible real-valued actions, dubbed proto-actions, in $\mathbb{R}^{m\times N}$. In particular, DDPG is an actor-critic algorithm, in which a critic network $Q_\theta(s, \hat{A})$ and an actor or policy network $f_\phi(s)$ are learned, such that $Q_\theta(s, \hat{A})$ can be evaluated at any $\hat{A} \in \mathbb{R}^{m\times N}$ and $f_\phi(s) \in \mathbb{R}^{m\times N}$, s being the user state, is trained to output a real-valued action with high Q-value:

$$f_\phi : \mathcal{S} \to \mathbb{R}^{m\times N}, \quad f_\phi(s) = \hat{A}\,. \tag{1}$$

Here, we write $\hat{A} \in \mathbb{R}^{m\times N}$ to distinguish from the ordered item set A. All value functions can be evaluated over any $\hat{A} \in \mathbb{R}^{m\times N}$, as well as at any slate A, by mapping items to their m-dimensional embeddings. In the *Proto-Slate* architecture, the Q-value of a slate is decomposed as

$$Q_\theta(s, \hat{A}) = \sum_{i=1}^{N} P(\hat{A}[i] \mid s, \hat{A})\overline{Q}_\theta(s, \hat{A}[i]) \tag{2}$$

where $\overline{Q}_\theta(s, \hat{A}[i])$ is an itemwise Q-value and $P(\hat{A}[i] \mid s, \hat{A})$ is the normalized user choice probability evaluated over the item embedding, of a user in state s selecting the item i when presented a slate $\hat{A}$, learned through a user choice model [14].

Essentially, the algorithm involves two optimization steps, carried out over a mini-batch $\mathcal{B} = \{(s, j, A, r, s')\}$ of tuples. Here, j is the item selected by the user when presented a slate A, s' is the transitioned state [13, 14] and r is the satisfaction reward from the environment [13, 14]. The parameters θ of the critic network are updated using a variant of the Slate-Q algorithm along with $f_\phi(s)$ to identify candidate items, see [13, 15]. In practice, as standard for DDPG, to improve training stability, two target networks are used. We omit this detail for clarity of presentation.

The parameters ϕ of the actor network are updated in a gradient descent step, optimizing the policy loss :

$$L(\phi) = -\sum_{s\in\mathcal{B}} \sum_{i=1}^{N} P(f_\phi(s)[i] \mid s, f_\phi(s))\overline{Q}_\theta(s, f_\phi(s)[i]) \tag{3}$$

In this work, we extend the Proto-Slate framework as shown in Figure 1. Slates are served to users in the simulation environment, which uses a trained user model to select an item from the slate, generate the reward and update the user state. To generate the replay buffer, a slate A is recommended by the system and then modified by a language model to A^{LLM}, using the same prompt as

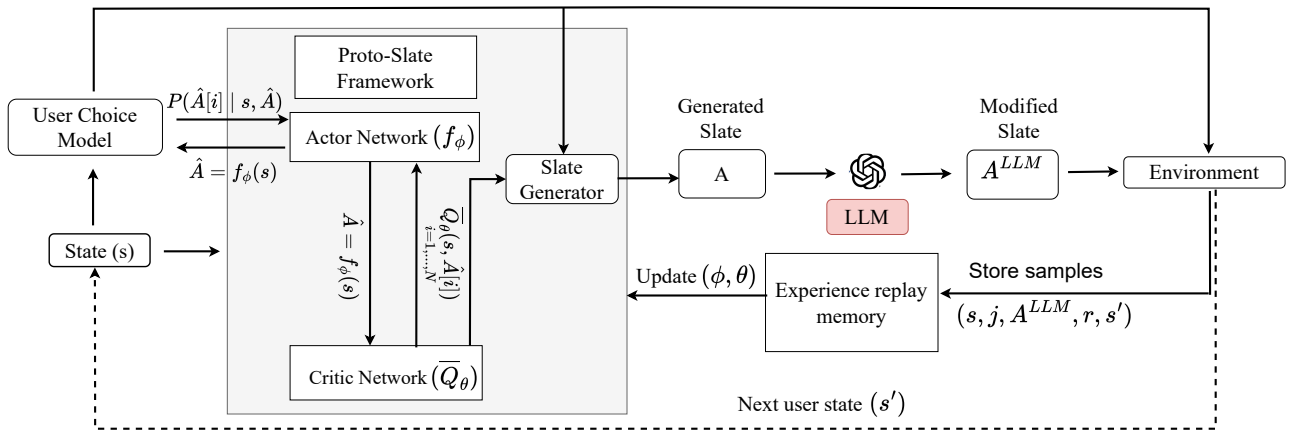

Figure 1: Overview of SlateLLM Training Framework. For Proto-Slate+LLM and SlateQ+LLM only the LLM component is used on top of their slate generation technique during inference.

mentioned earlier, before presenting it to the environment. Hence the training data consists of tuples of the form (s, j, A^{LLM}, r, s'). We can consider A^{LLM} to be a relatively high-quality action, closer to the optimal action than A [6, 9]. It makes sense then to regularize the policy loss, to encourage the learning of $f_\phi(s)$ towards slates close to those produced by the LLM [10].

To incorporate LLM knowledge into the existing slate recommendation framework, we modify the loss function of the actor network as follows:

$$\mathcal{L}(\phi) = -\sum_{s \in \mathcal{B}} \left(\lambda L_a + (1-\lambda) L_d\right) \tag{4}$$

where $\lambda \in [0, 1]$ controls the trade-off between maximizing Q-values and aligning with LLM-driven semantic supervision. The term L_a corresponds to the actor objective, defined in Equation 3, which encourages the selection of actions with high expected return. To improve numerical stability during training, we apply the tanh function to clip Q-values preventing them from exploding. The formulation is given by:

$$L_a = -\sum_{i=1}^{N} P(\hat{A}[i] \mid s, \hat{A}) \cdot \tanh\left(\overline{Q}_\theta(s, \hat{A}[i])\right) \tag{5}$$

where $\hat{A} = f_\phi(s)$. To guide the actor network f toward generating slates that reflect LLM-style semantic reasoning, we introduce a regularization term L_d based on action alignment [10]. In particular, we use the cosine similarity:

$$L_d = 1 - \frac{1}{N} \sum_{i=1}^{N} \cos\left(\hat{A}[i], A^{\text{LLM}}[i]\right) . \tag{6}$$

This encourages the learned proto-slate to semantically align with the guidance provided by the LLM, while still optimizing for long-term reward. We refer to this LLM-regularized variant of the Proto-Slate framework as SlateLLM, which captures LLM-style semantic reasoning during training while maintaining inference-time efficiency, making it well-suited for real-world deployment scenarios.

2.2 Offline Evaluation Paradigm

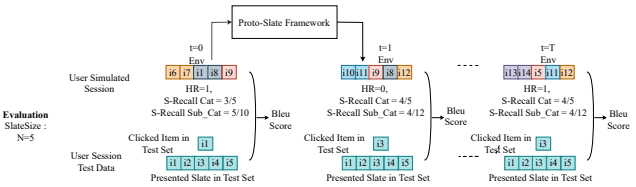

Figure 2: Evaluation example of a test user session through simulation environment

To evaluate our proposed policies, we use the publicly available MIND dataset [17] within a customized RecSim environment [7] built on PyTorch. The dataset provides session-level interaction data, including user click histories, presented news articles, and binary click indicators. Following [14], we preprocess multi-click impressions into single-click interactions so that each slate corresponds to one user choice. The simulator computes user state embeddings from historical data and generates slates iteratively across a session, capturing both immediate and long-term engagement. Evaluating offline RL policies on held-out data can suffer from selection bias, as better action sequences proposed by the learned policy might not appear in the test data [2]. To address this, we adopt a simulator-based evaluation paradigm, following the alternatives suggested by Deffayet et al. [2]. Specifically, inspired by RecSim [7] and Proto-Slate [14], we build a semi-synthetic simulator grounded in real-world interaction data, where at each step, a slate is generated by the policy under evaluation, and a learned user model selects an item from the slate to simulate user feedback. This feedback is then used to update the user state and generate subsequent slates, allowing for dynamic multi-step interactions rather than relying on static logged clicks. To further reduce overfitting to historical behavior, we introduce stochasticity by sampling user responses using a softmax distribution over item scores. While we do not apply full domain randomization, the stochastic modeling helps approximate user variability, narrowing the gap between

simulation and real-world dynamics as recommended by recent best practices [2]. As illustrated in Figure 2, for a test user u who clicks on items i_1, i_3, i_5, the session length is $T = 3$. At each time step, a new slate (e.g., $\{i_6, i_7, i_1, i_8, i_9\}$) is generated based on the current state. If the clicked item is present in the slate, the session progresses accordingly; otherwise, the trained user model selects an item.

Metrics. We evaluate slate quality using four metrics. Hit Ratio measures accuracy, averaged over the session for a long-term assessment. S-Recall [18] quantifies diversity at both the category and sub-category levels: category-level S-Recall counts the distinct categories among the 10 recommended items, while sub-category-level S-Recall computes the proportion of subcategories present relative to the overall subcategories present in the slate's categories. Novelty is measured as the average self-information [19] of the slate items, computed using a dampened inverse-frequency formulation:

$$\text{Novelty}(i) = \log_2\left(1 + \frac{1}{\text{u_count}(i) \times \log_2(1 + \text{t_clicks}(i))}\right) \quad (7)$$

where $\text{u_count}(i)$ denotes the number of users who interacted with item i, and $\text{t_clicks}(i)$ denotes the total number of clicks received by item i. This formulation assigns higher novelty scores to less popular items, encouraging exposure to rarer content. Additionally, we compute BLEU scores at the title level to assess lexical alignment between recommended slates and actual user preferences, providing a finer-grained measure of relevance. Evaluation metrics are averaged over T steps. This simulation-based approach enables reproducible and realistic evaluation of session-based policies without relying on costly online experiments.

3 Experiments & Results

We evaluate the slate construction methods by comparing the news articles included in the generated slates of 10 items with the actual time-ordered clicks recorded during each test user's session. The evaluation focuses on session-based slate algorithms that assume a single click per slate. Each training strategy is run on $16K$ user sessions, while SlateLLM requires $22K$ sessions to converge. We evaluate on a held-out test set comprising the $2K$ most recent user session trajectories, selected based on timestamp to better reflect real-world deployment conditions as opposed to random sampling. Across all experiments, we use the Adam optimizer with a learning rate of 0.001 for both Q-networks and policy networks, a Polyak averaging parameter [11] of $\tau = 0.0001$, and a balancing parameter $\lambda = 0.5$ for SlateLLM. We acknowledge LLMs can hallucinate and be prompt-sensitive; to ensure consistent, reproducible outputs, we fix the decoding temperature to 0, using the exact prompt stated in Section 2.

To analyze user behavior, we categorize users into generalists and specialists based on an entropy-based diversity metric computed from their click histories. For each user, we derive a probability distribution over news categories, calculate the entropy, and normalize it by the maximum possible entropy, yielding a diversity score between 0 and 1. Users in the first quartile are labeled as specialists, while the remainder are treated as generalists. Table 1 compares algorithmic performance across these groups. Although Proto-Slate+LLM shows marginally higher absolute scores, SlateLLM matches its performance without statistically significant differences while avoiding the infeasibility of using LLMs at inference. For novelty, Proto-Slate, Proto-Slate+LLM, and SlateLLM all outperform other methods, highlighting that the Proto-Slate architecture inherently enhances novelty. Additionally, LLM-based methods yield higher BLEU scores, reflecting better semantic alignment. Overall, as shown in Table 1, SlateLLM encapsulates the strengths of Proto-Slate's efficiency and the semantic richness of LLMs, delivering comparable performance across all metrics while maintaining practical deployability by eliminating the need for LLM inference in real-world serving scenarios.

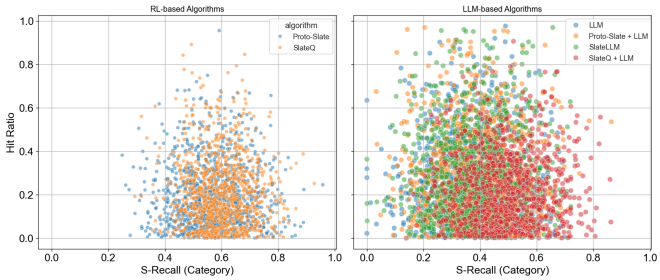

Figure 3: Relationship between Hit Ratio and S-Recall (Category) across models. Only users achieving positive hit ratios across all models are shown.

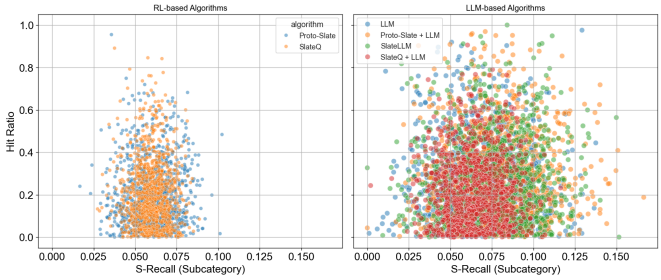

Figure 4: Relationship between Hit Ratio and S-Recall (Sub-Category) across models. Only users achieving positive hit ratios across all models are shown.

To analyze how slate recommendation models trade off between accuracy and diversity, we visualize the relationship between hit ratio and S-Recall across categories (Figure 3) and subcategories (Figure 4). We observe that LLM-influenced algorithms tend to reduce diversity at the broader category level while enhancing it at the finer subcategory level, as also reflected in Table 1. This suggests that LLMs promote fine-grained personalization, narrowing the topical spread of slates while deepening specificity. Such behavior may improve engagement by aligning recommendations more closely with nuanced user interests, albeit at the cost of broader topical exposure. SlateLLM, notably, manages to replicate these desirable diversity patterns without relying on LLMs at inference time, thereby retaining both interpretability and deployment efficiency. Furthermore, its consistent performance gains over Proto-Slate and SlateQ underline its potential as a scalable alternative that

Table 1: Performance Comparison of models for Generalist and Specialist Users. † indicates a statistically significant difference from Proto-Slate+LLM using paired Student's t-tests ($p \leq 0.05$). The absence of † for SlateLLM shows no significant difference, supporting its practical scalability. For reliable statistical analysis, each of the 3 seeded trained model runs is tested on the same user session trajectories, confirmed to be normally distributed and with equal variances by Shapiro-Wilk and Levene tests (95% confidence).

	Generalist					Specialist				
Algorithm	Hit Ratio	BLEU	S-Rec (C)	S-Rec (SC)	Nov.	Hit Ratio	BLEU	S-Rec (C)	S-Rec (SC)	Nov.
Proto-Slate	0.1559 ± 0.024†	0.3552 ± 0.011†	0.5845 ± 0.017†	0.0592 ± 0.008†	**0.6321** ± 0.013†	0.1427 ± 0.012†	0.3722 ± 0.015†	0.5677 ± 0.009†	0.0614 ± 0.010†	**0.6451** ± 0.014†
SlateQ	0.1481 ± 0.017†	0.3723 ± 0.010†	**0.6168** ± 0.018†	0.0602 ± 0.009†	0.5067 ± 0.011†	0.1467 ± 0.014†	0.3657 ± 0.013†	**0.6146** ± 0.016†	0.0588 ± 0.010†	0.5193 ± 0.015†
Proto-Slate+Sim	0.1495 ± 0.011†	0.3655 ± 0.009†	0.5425 ± 0.015†	0.0573 ± 0.007†	0.6261 ± 0.012†	0.1293 ± 0.013†	0.1946 ± 0.020†	0.5167 ± 0.014†	0.0594 ± 0.009†	0.6283 ± 0.011
SlateQ+Sim	0.1447 ± 0.010†	0.3807 ± 0.013†	0.5832 ± 0.017†	0.0587 ± 0.008†	0.5135 ± 0.014†	0.1449 ± 0.004†	0.4019 ± 0.011†	0.5709 ± 0.019†	0.0549 ± 0.007†	0.5305 ± 0.013†
LLM	0.1528 ± 0.021†	0.4833 ± 0.014	0.3696 ± 0.011†	0.0726 ± 0.009	0.6113 ± 0.012	0.1447 ± 0.009†	0.4860 ± 0.010†	0.3498 ± 0.014	0.0690 ± 0.008	0.5897 ± 0.015
Proto-Slate+LLM	**0.1903** ± 0.008	**0.4932** ± 0.009	0.3864 ± 0.013	**0.0780** ± 0.007	0.6186 ± 0.012	0.1593 ± 0.006	**0.5577** ± 0.008	0.3316 ± 0.012	**0.0704** ± 0.008	0.6075 ± 0.010
SlateQ+LLM	0.1708 ± 0.012†	0.4450 ± 0.010†	0.5545 ± 0.016†	0.0648 ± 0.008†	0.4839 ± 0.013†	0.1553 ± 0.011	0.5181 ± 0.014	0.3491 ± 0.013	0.0667 ± 0.009†	0.5340 ± 0.012†
SlateLLM	0.1835 ± 0.011	0.4857 ± 0.008	0.4216 ± 0.012	0.0742 ± 0.007	0.6285 ± 0.011	**0.1618** ± 0.007	0.5024 ± 0.009	0.3926 ± 0.013	0.0695 ± 0.008	0.6310 ± 0.010

balances accuracy with personalized diversity. Code is available in GitHub.[1]

4 Conclusion & Future Work

This work presents an initial exploration into incorporating LLM-based semantics into session-based slate recommendation. We propose SlateLLM, a model that learns the fine-grained personalization capabilities of LLM reasoning while preserving the novelty-promoting characteristics of existing frameworks like Proto-Slate. By introducing a regularization strategy during training, SlateLLM distills LLM-guided slate refinements without requiring LLM inference at serving time, offering a scalable and deployable solution. While training demands increase, our evaluation, unlike prior simulation only approaches, leverages a real-world data-driven simulator with multi-slate interactions to assess long-term engagement. Results show that SlateLLM improves accuracy, diversity, and novelty, demonstrating its potential for practical, real-world recommender systems. Future work will explore adaptive regularization techniques and lightweight LLM distillation methods to further reduce training cost while preserving semantic alignment.

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